

Operator-Word Calculus on (Q, p) : Slides, Blow-Ups, Hyperbolic Stabilization, and Rewrite-to-NF(Q, p)

Ryan Van Gelder

2025

Abstract

We formalize a state-based rewriting logic for an operator-word calculus acting on configurations (Q, p) , where Q is an integral symmetric form and p is an integral refinement vector. The generator set consists of slides $S_{ij}^\pm$ (elementary unimodular basis changes), blow up/down moves $B_\pm, B_\pm^{-1}$, and hyperbolic stabilization/cancellation H, H^{-1} . A deterministic rewrite strategy produces a canonical normal form $\text{NF}(Q, p)$ that is invariant under the generated equivalences. We also indicate a specialization to 8D plumbings bounding Milnor 7-spheres, making admissibility tests and the canonical p -data explicit.

Background pointers. Exotic 7-spheres and Θ_7 : [11, 9]. Smoothing theory: [15]. Kirby calculus: [8, 7].

1 Configurations and semantics

1.1 Configuration space

A configuration is a pair

$$(Q, p), \quad Q \in \text{Sym}_n(\mathbb{Z}), \quad p \in \mathbb{Z}^n.$$

Interpretation: Q is a lattice pairing (e.g. an intersection form in a chosen basis) and p is extra structure (e.g. a characteristic/Pontryagin-refinement datum in that basis).

1.2 Congruence (“gauge”) action

A unimodular basis change $U \in \text{GL}_n(\mathbb{Z})$ acts by

$$(Q, p) \longmapsto (U^T Q U, U^{-1} p).$$

Slides are elementary generators for this action (handle slides in Kirby calculus; see [8, 7]).

2 Generator operators (raw moves)

2.1 Slides S_{ij}^ε

For $i \neq j$ and $\varepsilon \in \{+1, -1\}$, let

$$U_{ij}(\varepsilon) = I + \varepsilon E_{ij} \in \text{GL}_n(\mathbb{Z}),$$

where E_{ij} has a single 1 in position (i, j) .

Definition (Slide).

$$S_{ij}^\varepsilon(Q, p) = (U_{ij}(\varepsilon)^T Q U_{ij}(\varepsilon), U_{ij}(\varepsilon)^{-1} p).$$

Slides are total and invertible: $(S_{ij}^\varepsilon)^{-1} = S_{ij}^{-\varepsilon}$.

2.2 Blow-up/down B_σ

For $\sigma \in \{+1, -1\}$ define the blow-up

$$B_\sigma(Q, p) = (Q \oplus [\sigma], p \oplus 0).$$

This is total.

Blow-down B_σ^{-1} (partial). It is admissible only if (Q, p) contains an orthogonal primitive σ -vector (precisely in §3.3).

2.3 Hyperbolic stabilization/cancellation H

Let $H_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Define

$$H(Q, p) = (Q \oplus H_2, p \oplus (0, 0)).$$

Cancellation H^{-1} is partial: it is admissible only if (Q, p) contains an orthogonal hyperbolic plane (precisely in §3.5).

3 Rewriting logic: explicit rules on configurations

We write rewrite rules as

$$\langle Q, p \rangle \Rightarrow \langle Q', p' \rangle$$

possibly under an admissibility predicate.

3.1 Slide rule (always admissible)

$$(\text{SLIDE}) \quad \overline{\langle Q, p \rangle \Rightarrow \langle U^T Q U, U^{-1} p \rangle}$$

where $U = U_{ij}(\varepsilon) = I + \varepsilon E_{ij}$.

3.2 Blow-up rule (always admissible)

$$(\text{BLOWUP}) \quad \overline{\langle Q, p \rangle \Rightarrow \langle Q \oplus [\sigma], p \oplus 0 \rangle} \quad \sigma \in \{\pm 1\}.$$

3.3 Blow-down rule (partial)

We formalize blow-down by finding a unimodular U such that

$$U^T Q U = Q' \oplus [\sigma].$$

Let π be projection onto the first $n - 1$ coordinates.

Predicate $\text{Split}_\sigma(Q)$. There exists primitive $v \in \mathbb{Z}^n$ and $U \in \text{GL}_n(\mathbb{Z})$ with $Ue_n = v$ and $U^TQU = Q' \oplus [\sigma]$.

$$(\text{BLOWDOWN}) \quad \frac{\text{Split}_\sigma(Q) \wedge U^TQU = Q' \oplus [\sigma]}{\langle Q, p \rangle \Rightarrow \langle Q', \pi(U^{-1}p) \rangle}$$

In rewriting-to-normal-form we do not allow arbitrary U ; we choose a canonical witness (see §4.1).

3.4 Hyperbolic add rule (always admissible)

$$(\text{HYPADD}) \quad \overline{\langle Q, p \rangle \Rightarrow \langle Q \oplus H_2, p \oplus (0, 0) \rangle}.$$

3.5 Hyperbolic cancel rule (partial)

Predicate $\text{HypSplit}(Q)$. There exist primitive $u, v \in \mathbb{Z}^n$ and $U \in \text{GL}_n(\mathbb{Z})$ so that $Ue_{n-1} = u$, $Ue_n = v$, and

$$U^TQU = Q' \oplus H_2.$$

Equivalently, in the original basis: $u^TQu = 0$, $v^TQv = 0$, $u^TQv = 1$, and the span splits off orthogonally.

Let π project away the last two coordinates. Then

$$(\text{HYPCANCEL}) \quad \frac{\text{HypSplit}(Q) \wedge U^TQU = Q' \oplus H_2}{\langle Q, p \rangle \Rightarrow \langle Q', \pi(U^{-1}p) \rangle}.$$

Again, normalization uses a canonical choice of (u, v, U) (see §4.1).

4 Deterministic rewrite-to-normal-form strategy

The central idea is: *words are not normalized syntactically; states are normalized deterministically.*

Define a deterministic normalization map

$$\mathbf{N} : (Q, p) \longmapsto \text{NF}(Q, p).$$

We obtain a normalized step relation

$$(Q, p) \xrightarrow{g} \mathbf{N}(g(Q, p))$$

for any generator g admissible at (Q, p) .

4.1 Canonical witness selection for partial reductions

To make $\mathbf{N}$ deterministic, every time a partial rule (BLOWDOWN/HYPCANCEL) is applicable, we choose the witness by a fixed minimization procedure.

Canonical blow-down choice

Among all primitive v that split off a $[\sigma]$, select the one minimizing

$$(\|v\|_2, \text{lex}(v)).$$

Then among unimodular U with $Ue_n = v$ and $U^TQU = Q' \oplus [\sigma]$, choose U minimizing a deterministic tie-break key (e.g. lexicographic order of the last row, then of the full matrix entries).

Canonical hyperbolic cancellation choice

Among all primitive hyperbolic pairs (u, v) that split off H_2 , select the one minimizing

$$(\max(\|u\|_2, \|v\|_2), \text{lex}(u), \text{lex}(v)).$$

Then choose a canonical completing U as above.

Stage N1 — Stabilization reduction.

1. While a canonical blow-down is admissible, apply (BLOWDOWN) using the canonical witness.
2. While a canonical hyperbolic cancellation is admissible, apply (HYPCANCEL) using the canonical witness.
3. Repeat until neither applies.

This strictly decreases rank whenever it fires, hence terminates.

Stage N2 — Global congruence normalization (real-place basis fix). Apply a deterministic lattice reduction procedure to fix the remaining congruence freedom, e.g.:

- compute an LLL-reduced basis for the lattice with Gram matrix Q [10],
- among all LLL-valid reduced bases, choose the one that lexicographically minimizes the coordinates of p ,
- fix a sign convention (first nonzero entry of p positive; then lex-min for remaining symmetries).

This yields a reproducible representative in the congruence class.

Stage N3 — Prime-local block normalization (optional). (Local classification/Jordan decompositions and invariants of integral quadratic forms: see [12, 13, 14].) For each selected prime ℓ (and optionally depth r):

1. Compute the ℓ -adic Jordan decomposition of $(Q, p) \otimes \mathbb{Z}_\ell$:

$$(Q, p) \otimes \mathbb{Z}_\ell \cong \bigoplus_{e \geq 0} \ell^e (Q_{\ell, e}, p_{\ell, e}).$$

2. If using a projector Π_{ℓ^r} , truncate to $0 \leq e < r$.
3. Sort blocks by a canonical key, e.g.

$$(e, \text{rank}, \det \bmod (\mathbb{Z}_\ell^\times)^2, \text{Hasse/2-adic refinements}, \text{coordinates of } p_{\ell, e}).$$

The result is $\text{NF}(Q, p)$. If only a stable normal form is needed, one may stop at Stage N2.

5 Operator words, evaluation, and equivalence

5.1 Words

A word is a sequence of generator symbols

$$w = g_m \cdots g_2 g_1$$

subject to admissibility of partial inverses at runtime.

5.2 Normalized evaluation

Define

$$\text{EvalNF}(w; Q, p) := \mathbf{N}(w(Q, p)).$$

Two words are equivalent on input (Q, p) if they yield the same normal form.

5.3 “Resonance = 1” criterion (move invariance)

Replace any scoring notion by strict equality of canonical outputs:

$$\text{EvalNF}(w_1; Q, p) = \text{EvalNF}(w_2; Q, p).$$

6 Specialization: 8D plumbing coboundaries of Milnor 7-spheres

This section makes (Q, p) explicitly computable from an 8D plumbing of disk bundles over S^4 whose boundary is a homotopy 7-sphere (Milnor-type).

6.1 Plumbing input

A plumbing description consists of:

- a finite graph $\Gamma = (V, E)$;
- for each vertex $v \in V$, an oriented D^4 -bundle over S^4 with structure group $SO(4)$, specified by bundle parameters $(a_v, b_v) \in \mathbb{Z} \oplus \mathbb{Z}$ (a standard $\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$ coordinate choice; see [5, 4]);
- each edge represents a plumbing along fiber/base disks with a chosen sign convention.

Let $n = |V|$ and choose the basis $\{x_v\}_{v \in V} \subset H^4(W; \mathbb{Z})$ given by the core S^4 of each bundle.

6.2 Constructing the intersection matrix Q

Define $Q \in \text{Sym}_n(\mathbb{Z})$ by:

- Diagonal:

$$Q_{vv} = e_v, \quad e_v := a_v + b_v \quad (\text{Euler number in this convention}).$$

- Off-diagonal: for $v \neq w$,

$$Q_{vw} = \sum_{\text{edges } (v,w)} \epsilon_{vw},$$

where $\epsilon_{vw} \in \{\pm 1\}$ is the plumbing sign (often all +1 in standard Milnor plumbings).

This is the usual “diagonal = self-intersection, edges = intersection” plumbing matrix pattern, in degree 4 for an 8-manifold.

6.3 Constructing the refinement vector p

In the same basis $\{x_v\}$ define $p \in \mathbb{Z}^n$ by

$$(p)_v := a_v - b_v \quad (\text{a standard choice compatible with } p_1 \text{ conventions up to overall factors}).$$

Spin constraint / admissibility. To be in the spin setting, require the appropriate mod 2 parity condition on the vertex data (equivalently $w_2(W) = 0$); in many plumbing conventions this forces each Euler number e_v to be even.

6.4 A boundary scalar from (Q, p) (Eells–Kuiper type)

Once (Q, p) is formed for the coboundary W^8 , define:

- Signature $\sigma(W) = \text{sign}(Q)$;
- Quadratic value (over $\mathbb{Q}$):

$$p_W^2 := p^T Q^{-1} p \in \mathbb{Q}.$$

Then a boundary invariant (for homotopy 7-spheres in the relevant class) is built from

$$\mu \equiv \frac{p_W^2 - \sigma(W)}{224} \pmod{1},$$

up to convention choices (Eells–Kuiper invariant [6]; modern classification context [3]). This makes the 2- and 7-primary structure explicit via the denominator.

6.5 Admissibility tests in plumbing cases (explicit)

In plumbing-derived Q , admissibility for cancellations often becomes concrete.

Blow-down test (± 1 -sphere split). Search for primitive v with

$$v^T Q v = \pm 1.$$

If found, attempt to complete v to a unimodular basis and check if the last basis vector is orthogonal to the others (i.e. off-diagonal last row/column can be eliminated by slides).

Hyperbolic split test. Search for primitive u, v with

$$u^T Q u = 0, \quad v^T Q v = 0, \quad u^T Q v = 1,$$

and verify orthogonal split by completing (u, v) to a unimodular basis and reducing cross-terms via slides.

These tests are exactly what Stage N1 requires.

7 Compact rewriting-logic summary

7.1 Rule schema list

- (SLIDE) $\langle Q, p \rangle \Rightarrow \langle U^T Q U, U^{-1} p \rangle$ for $U = I \pm E_{ij}$.
- (BLOWUP) $\langle Q, p \rangle \Rightarrow \langle Q \oplus [\sigma], p \oplus 0 \rangle$.
- (BLOWDOWN) $\langle Q, p \rangle \Rightarrow \langle Q', \pi(U^{-1} p) \rangle$ if $U^T Q U = Q' \oplus [\sigma]$ (canonical witness).
- (HYPADD) $\langle Q, p \rangle \Rightarrow \langle Q \oplus H_2, p \oplus (0, 0) \rangle$.
- (HYPCANCEL) $\langle Q, p \rangle \Rightarrow \langle Q', \pi(U^{-1} p) \rangle$ if $U^T Q U = Q' \oplus H_2$ (canonical witness).

7.2 Deterministic strategy

Define $\text{NF}(Q, p) = \text{N}(Q, p)$ by:

1. Apply canonical (BLOWDOWN)/(HYPCANCEL) until stuck.
2. Apply deterministic congruence normalization (LLL + lex tie-break on p).
3. Optionally apply prime-local block normalization (Jordan decomposition + sorted blocks), including truncations Π_{ℓ^r} when desired.

8 Suggested minimum viable experiment (Milnor 7-spheres)

Input. A plumbing graph Γ with vertex parameters (a_v, b_v) and edge signs.

Compute.

1. (Q, p) via §6.
2. $\text{NF}(Q, p)$ via §4.
3. Prime-tier fingerprints via Stage N3 (especially $\ell = 2$ and $\ell = 7$).
4. Scalar μ and its 2- and 7-primary projections.

Validate invariance. Randomly apply sequences of slides (and optional stabilizations followed by normalization) and confirm $\text{NF}(Q, p)$ is unchanged.

9 Notes on conventions

The precise mapping from vertex parameters (a_v, b_v) to $(e_v, (p)_v)$ depends on orientation/normalization conventions for $SO(4)$ -bundles over S^4 . The rewrite calculus is invariant under any consistent choice, as long as:

1. Q is integral symmetric and transforms by congruence under slides,
2. p transforms as $p \mapsto U^{-1} p$, and
3. stabilizations append zeros to p .

If you adopt a different convention, record it once and keep it fixed; $\text{NF}(Q, p)$ remains well-defined for that convention.

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